

SARAH MADISON TAYLOR

West Palm Beach, Florida | mtsarah016@gmail.com | 205-533-3470

RESEARCH INTERESTS

Computational environmental science; climate change and environmental modeling; computational approaches to disease and public health; spatial analysis and GIS; ecological modeling; agent-based modeling; data science; climate impacts on disease systems; and interdisciplinary applications of computer science to biological research.

EDUCATION

Rhodes College — Memphis, Tennessee

Bachelor of Science, Double Major in Computer Science and Environmental Science | May 2026

RESEARCH PROJECTS

Adaptive Resource Allocation Under Climate Change —

***Using Computational Modeling with Malaria Vaccination Coverage as a Case Study* |**

- Investigated computational approaches to malaria vaccination resource allocation under changing climate conditions.
- Integrated environmental and epidemiological datasets using Python, GIS, and computational modeling.
- Applied spatial and computational approaches to examine relationships among climate conditions, disease risk, and vaccination coverage.
- Explored resource-allocation strategies under projected environmental conditions.

Agent-Based Modeling Applied to Weevil Predation on the Endangered *Tillandsia utriculata* Species —

***Computational Ecological Modeling* |**

- Developed agent-based models using NetLogo to investigate ecological interactions involving weevil predation.
- Used computational simulation to examine population-level ecological outcomes.
- Applied concepts from ecology, environmental science, and computer science to model an endangered species system.

PRESENTATIONS

Taylor, S. (2026). *Adaptive Resource Allocation Under Climate Change: Using Computational Modeling with Malaria Vaccination Coverage as a Case Study*. Rhodes Symposium, Rhodes College, Memphis, TN.

Taylor, S. (2026). *Bloom* [Software Engineering Live Demo]. Rhodes Symposium, Rhodes College, Memphis, TN.

Taylor, S. (2024). *Mayan Colonization in Belize*. Tropical Research and Education Center, San Pedro, Belize. Rhodes College Maymester: Marine Biology and Climate Change.

Taylor, S. (2024). *Renewable Energy in Belize*. Tropical Research and Education Center, San Pedro, Belize. Rhodes College Maymester: Marine Biology and Climate Change.

PROFESSIONAL EXPERIENCE

The Herbert Wertheim UF Scripps Institute for Biomedical Innovation & Technology — Palm Beach, Florida

Research Technician, Yuan Lab | August 2026–Present

- Contribute to laboratory research and daily research operations within a biomedical research laboratory.
- Collect, maintain, and document research data and biological specimens according to established laboratory protocols.
- Assist investigators and senior laboratory personnel with preliminary data analysis, research documentation, summary reports, and presentation of findings.
- Support physiological and cellular research experiments involving digital imaging, fluorescence microscopy, and ultrasound-based measurements.
- Operate and maintain laboratory instrumentation including microscopes, fluorescence microscopes, centrifuges, analytical balances, cell incubators, and cell/tissue storage equipment.
- Organize scientific materials, laboratory supplies, and research equipment and assist with procurement and inventory management.
- Support equipment maintenance and coordinate repairs when necessary.
- Collaborate with the Animal Research Compliance team to support protocol compliance.
- Mentor students and assist with laboratory meetings, scheduling, agendas, and research-related administrative activities.

Royal Caribbean — Alaska

Assistant to Ship Naturalist | June 2026

- Researched and developed a presentation on seals and sea lions for the ship's naturalist, to deliver to passengers during Alaskan cruises.
- Synthesized ecological and behavioral information on seal and sea lion species into passenger-accessible content.
- Provided passengers with basic technical support

Rhodes College Environmental Science and Studies Department — Memphis, Tennessee

Rhodes Student Associate | August 2025–December 2025

- Supported faculty research and academic programming within the Environmental Studies Department.
- Organized research talks and discussions involving students and faculty.
- Assisted with communication and coordination of departmental research opportunities.
- Served as a departmental teaching assistant and communicated academic and research opportunities to students.
- Worked with faculty to gather and organize information related to current research activities and courses.

Memphis Zoo — Memphis, Tennessee

Research Technician Intern | October 2025–May 2026

- Participated in conservation and research activities involving endangered Louisiana pinesnakes and dusky gopher frogs.
- Provided daily animal care and monitored animal health within a high-biosecurity research environment.
- Assisted researchers and veterinarians during biological sample collection, ultrasounds, and health assessments.
- Collected, documented, and maintained biological samples and animal health information.

Town of Atoka — Atoka, Tennessee

GIS Intern | January 2026–May 2026

- Collected and managed geographic data using GPS, mobile applications, field verification, and GIS tools.
- Analyzed municipal GIS needs and developed GIS-based solutions for public works and other Town departments.
- Designed and produced maps using ArcGIS and ArcGIS Online for municipal departments, elected officials, and external agencies.
- Worked with spatial datasets representing municipal infrastructure including water boundaries, hydrants, sewer systems, water tanks, and valves.

Rhodes College Finance Office — Memphis, Tennessee

Work Study Student | January 2023–May 2026

- Organized, scanned, filed, and maintained confidential financial and administrative documentation.
- Maintained organized digital and physical filing systems to improve accessibility and office efficiency.
- Handled sensitive information professionally and followed established procedures for confidential records.
- Communicated with students, faculty, and staff while providing general administrative support.

TECHNICAL & LABORATORY SKILLS

Programming: Python, Java, C, R, NetLogo, Git/GitHub

Computational & Data: Computational modeling, agent-based modeling, data analysis, spatial analysis

GIS: ArcGIS, ArcGIS Online, GPS data collection, spatial data management

Laboratory: Biological specimen collection and documentation, microscopy, fluorescence microscopy, digital imaging, centrifugation, analytical balances, cell incubators, cell/tissue storage

Research: Scientific documentation, data management, research presentations, literature/research support

HONORS & ACADEMIC ACTIVITIES

- Beta Beta Beta (TriBeta) — Biology Honor Society
- Kappa Alpha Omicron — Environmental Science Honor Society
- Rhodes Journal of Biological Science — Editorial Staff
- Rhodes College Beekeeping Club — Secretary
- Alpha Omicron Pi — Director of Panhellenic Relations

INTERNATIONAL & STUDY ABROAD EXPERIENCE

- Science of Climate Change — Belize & Guatemala (2024)
- Web Applications — Czech Republic (2025)
- Computer Networking — South Korea (2025)